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ORIGINAL PAPER

Invading brown bullhead catfish (*Ameiurus nebulosus*) show thermal niche rigidity and benthic niche plasticity: implications for management

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Abstract Species invasions remain a key contributing factor to the global rise in extinction rates of native fishes in freshwater ecosystems. While efforts to remove non-native species from newly invaded water bodies often target well-established ecological characteristics and habitat associations, invasive species may exhibit different behaviour to conspecifics in their native range. Brown bullhead catfish (*Ameiurus nebulosus*) is a widespread invader characterized in their native range as a thermophilic nocturnal littoral-benthic predator with high site fidelity. We used acoustic telemetry to examine seasonal movement patterns and habitat use of catfish in Lake Rotoiti, a recently invaded lake in Aotearoa-New Zealand. Invasive catfish demonstrated strong habitat niche plasticity and thermal niche rigidity. Only in spring and summer were catfish regularly found in nearshore areas (between shore and < 7 m bottom depth). Instead, catfish were highly mobile, especially in autumn when catfish routinely occupied offshore regions (≥ 7 m bottom depth; 80% of positions), where they rarely associated with the lake bottom. Invasive catfish sought the warmest water temperatures available in the lake and their movement to offshore areas in autumn, along with marked increases in movement rates and nightly areas of use at this time, coincided with warmer waters in the offshore region.

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Informed by best practice, much of the catfish control effort in Lake Rotoiti relies on nearshore benthic netting. Our findings suggest that while this strategy may be effective in spring and summer, characterising and targeting thermal optimal habitat is a promising avenue.

Keywords Acoustic telemetry · Habitat · Movement · Niche · Aotearoa-New Zealand · Freshwater

Introduction

Extinction rates of native fishes continue to increase, with recent estimates that up to one-third of native freshwater fishes are at risk (Tickner et al. 2020; Sayer et al. 2025; Vardakas et al. 2025). The arrival and spread of invasive fish is considered one of the key factors responsible for biodiversity loss in freshwater ecosystems (Dudgeon et al. 2006; Vander Zanden et al. 2015). Many temperate freshwater systems have low levels of biodiversity, which provides increased opportunity for invasive species to enter and establish due to the naïveté of the receiving community (Carthey and Banks 2014). Overall low redundancy in species' roles associated with low freshwater biodiversity suggests that the introduction of a species with a distinct niche may also destabilize the ecosystem by rerouting trophic and nutrient pathways (Britton 2023).

There is an increasing body of knowledge showing that when species invade novel environments they can exhibit different behaviour and habitat choices relative to conspecifics within their native range (reviewed in Pyper et al. 2024). This may be in part driven by the effect of personality on invasion success by founder individuals (e.g., Brodin et al. 2013). Comparisons of species in their native versus invaded ranges also show that release from natively co-occurring enemies supports invasion success (Heger and Jeschke 2014), and this release may allow for a broader spectrum of behaviour and habitat use. Increased dispersal rates, exploratory behaviour, and trophic position at the individual or population level can give invaders a competitive advantage over native fishes, while simultaneously shifting ecosystem dynamics. The chance to inhabit novel freshwater environments compared to those encountered in their native range may relax the constraints on typical niche occupancy by the invader (Bates and Bertelsmeier 2021).

Understanding which aspects of an invasive species' niche are fixed and which are likely to expand within a novel environment is critical knowledge for modern, targeted incursion control strategies and for predicting the potential spread of new invaders across the landscape. After an invasive species has been detected, the subsequent incursion control is a critical stage (Sato et al. 2010; Rytwinski et al. 2019; Roberston et al. 2020; Ahmed et al. 2022) and effective control strategies commonly exploit knowledge of the invader's ecological niche (Venette et al. 2021). However, knowledge of the target species' ecological niche from a species' native range fails to account for niche expansion and behavioural flexibility within novel habitats. Management strategies can leverage novel invasion behaviours if they are identified and explicitly incorporated into incursion control.

Brown bullhead catfish (*Ameiurus nebulosus*; hereafter catfish) are native to North America, where they inhabit slow-moving water bodies, are associated with benthic habitats, have an optimum temperature of 25 °C (Hartman 2017), and have a broad omnivorous diet (Scott and Crossman 1973). This catfish has been introduced far outside their native range. In Europe they are known to negatively impact native freshwater fish communities, often becoming dominant soon after invasion (Rechulicz and Plaska 2021; Kalinowska et al. 2023). Catfish were introduced to Aotearoa-New Zealand (hereafter Aotearoa) by acclimatisation societies in 1877 (McDowall 1994). While originally confined to waterbodies near Auckland in the North Island, they eventually spread to other areas of the Island, including Lake Taupō (Australasia's largest lake) in the mid-1980s (Barnes and Hicks 2003). More recently catfish have been reported in multiple catchments in the central North Island (Hicks and Allan 2018). Like other members of the order Siluriformes, their feeding behaviour can promote sediment resuspension and thereby increase bioturbidity (Rowe 2007; Collier and Grainger 2015) which, along with broad temperature tolerances and omnivorous diets that include predation of many native species (Champion et al. 2002; Dedual 2019), creates a major threat to freshwater ecosystems (Cucherousset et al. 2018). Modelling studies demonstrate that catfish have a high potential to substantially expand their range in Aotearoa (Rowe and Wilding 2012; Collier and Granger 2015; Leathwick et al. 2016; Lee et al. 2025). This has serious implications for a range of unique native fauna with high conservation significance, including many taonga (culturally treasured) and mahinga kai (traditional harvest practice) species, which are considered integral to life by Māori (Barnes and Hicks 2003; Kusabs and Quinn 2009; Dedual 2019).

Telemetry is a powerful tool for quantifying spatio-temporal habitat use—and movement ecology more broadly—of invasive fishes (Lennox et al. 2016). For example, the habitat choice patterns of common carp (*Cyprinus carpio*) revealed by radio telemetry, enabled their successful eradication in an Australian lake (Taylor et al. 2012) and the study of goldfish (*Carassius auratus*) movements in an embayment of the Laurentian Great Lakes provided insight into their spawning sites and range expansion potential (Boston et al. 2024). Current catfish incursion control tends to target littoral zones within temperate lakes based on the assumption of high site fidelity (Sakaris et al. 2005a; Millard et al. 2009) and fixed benthic (Scott and Crossman 1973) and warmwater (Hartman 2017) niches. However, in larger, deeper lakes that thermally stratify during the summer months and do not freeze over winter, such as those recently invaded in Aotearoa, a benthic-littoral control strategy misses significant potential foraging habitat as it precludes the vast offshore areas. Furthermore, at certain times of the year, offshore habitat may be preferable as it is more thermally stable, and generally warmer, over the winter months (Woolway et al. 2022). Evidence of offshore habitat occupancy by invading catfish in freshwater lakes (Dedual 2002; Ulikowski et al. 2022) highlights the potential for expanding one or both of their thermal and benthic niches to take advantage of offshore foraging opportunities. Sustained measurements of catfish habitat use through acoustic telemetry will help elucidate their behaviour in a novel context.

Rotoiti i kite a Ihenga (hereafter Lake Rotoiti), a 3460-ha, deep mesotrophic lake in Rotorua in the Bay of Plenty region of central North Island, presents a case study for examining niche plasticity of invasive brown bullhead catfish. Catfish were detected in Lake Rotoiti in 2016, although incidental reports and age data from the invasive population indicate they were present in the lake several years prior to detection (Francis 2019; Kusabs et al. 2026). Following the catfish invasion of lakes in this region numbers have increased dramatically despite a netting programme that culled ~ 180,000 catfish over eight years, since 2016 (https://tearawa.io/programmes/). Catfish management efforts have largely focused on the use of fyke nets in the nearshore areas of Lake Rotoiti, and while this method results in high catch rates (Loette and Declerk 2006), the continued rapid population expansion highlights the need for a broader understanding of the spatial ecology and habitat use by this invader. To this end, we used spatial positioning acoustic telemetry that incorporated transmitters with pressure and temperature sensors to estimate three-dimensional catfish space use and thermal occupancy over a one-year period in the western basin of the lake; the area where the catfish invasion is believed to have started, and, at the time of the study, where abundance was highest (Francis 2019). Telemetry data were supported by high resolution modelled water temperature and lake bathymetry. Our aim was to quantify the daily and seasonal movements of catfish in a recently invaded lake to examine their realised thermal, benthic affiliation and niche dimensions, in comparison to the habitat use that would be anticipated in their native range. An additional goal is that our findings will be used to inform catfish management efforts, given the high invasion potential of catfish in Aotearoa and elsewhere.

Methods

Study area

Lake Rotoiti is a large, deep, monomictic lake that stratifies for nine months of the year, beginning in the spring and mixing once annually in late autumn (Gibbs 1992; Kusabs and Quinn 2009). The lake has two main basins, a shallower western basin ($Z_{max} = 32$ m) and a deeper eastern basin ($Z_{max} = 124$ m; Kusabs et al. 2015; Fig. 1). The shallower western basin, where this study was focused, has a higher degree of embayment than the eastern basin, and receives the main inflow (~ 80%) to the lake from the adjacent eutrophic Lake Rotorua through the Ohau channel. To help prevent declines in water quality in Lake Rotoiti, a 1.3 km wall diverts this inflow directly into the Okere Inlet and out

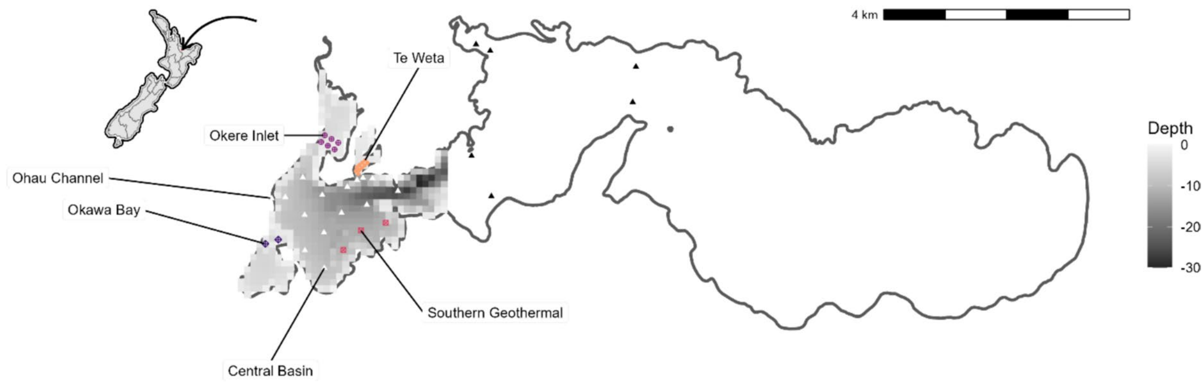

Fig. 1 Map of Lake Rotoiti, located on the North Island of Aotearoa New Zealand (shown in the top left inset) with acoustic telemetry receiver array shown. Spatial triangulation of tagged brown bullhead catfish in the western basin of the lake was possible with receivers indicated by orange solid circles (Te Weta), purple closed circles (Okere Inlet), pink squares (Southern Geothermal), and white triangles (central basin). Data from the "east" receivers (black triangles) were retained for temperature and depth analyses. Bathymetry data is shown on a 100 m^2 grid with depth measurements in metres

into the Kaituna River, the sole outflow to the lake (Fig. 1). The lake is classified as mesotrophic per 5-year mean (2017–2022), with total phosphorus, total nitrogen, chlorophyll *a*, and secchi depth values of 23.7 µg L^{-1}, 169.4 µg L^{-1}, 5.2 µg L^{-1}, and 6.2 m, respectively (Prentice et al. 2026).

The fish community in Lake Rotoiti is comprised of both natives and established non-natives as well as the recently invaded brown bullhead catfish. Native fishes include kōaro (*Galaxias brevipinnis*), common bully (*Gobiomorphus cotidianus*), and common smelt (*Retropinna retropinna*) (Kusabs and Quinn 2009). In the late 1800s, brown trout (*Salmo trutta*), rainbow trout (*Oncorhynchus mykiss*), and morihana (goldfish) were introduced (Burstall 1980 in Kusabs and Quinn 2009; Tadaki et al. 2025). None of these species overlap with brown bullhead catfish in their native range (Elliott 1989; MacCrimmon 1971; and Takada et al. 2010, respectively). Kōura (freshwater crayfish, *Paranephrops plainfrons*) are also present in the lake and like many native fish such as kōaro, they fulfil important cultural and ecological roles in Aotearoa lakes (Parkyn et al. 2001; Kusabs and Quinn 2009; Kusabs et al. 2015, 2026).

Fish telemetry

Catfish were captured in 4–6 mm mesh baited fyke nets set overnight in four nearshore regions of the shallow western bay of Lake Rotoiti (Okere Inlet, Te Weta Bay, Okawa Bay, and Southern Geothermal shoreline, Fig. 1; Francis 2019). A total of 30 catfish were selected for transmitter implantation, which took place over ~ 2-week period (Nov 20–Dec 6, 2018) in late spring. Catfish were anesthetized using AQUI-S (isoeugenol), measured (total length, nearest mm), weighed (nearest g), and then surgically implanted with acoustic transmitters (Vemco Ltd., Canada) either with temperature and pressure sensors (V9TP, n = 15; 9 mm diam × 31 mm) or without sensors (V7, n = 15; 7 mm diam × 19.5 mm) through a 1-cm incision into the abdominal cavity (Table 1). Both types of transmitters were programmed to have the same rate of transmission (random delay of 40–100 s). Each incision was closed using three absorbable sutures. Catfish were held overnight in cages before being released back into the general area of the lake where they were originally captured (Fig. 1, Table 1). All procedures were done in accordance with University of Waikato Animal Ethics Committee animal ethics permit (AEC-UOW-11033033).

A total of 42 omnidirectional acoustic hydrophone-receivers (VR2W or VR2Tx, 69 kHz, Vemco Ltd., Canada), each anchored ~ 2 m off the bottom of Lake Rotoiti, were deployed to monitor the movements of acoustically tagged catfish, as well as their depth and temperature occupancy, for a period of just over one year (Nov 2018–Dec 2019; Fig. 1). Based on the results of range tests carried out in May 2017, thirty-three receivers were spatially configured in the lake to

Table 1 Brown bullhead catfish were surgically implanted with acoustic transmitters before being released at the same capture location in Lake Rotoiti (see Fig. 1)

Tagging location	Sensor type	Number of fish	Fish mass (g)	Fish length (mm)	Detection duration (d)
Okawa Bay	None	5	143 (117–173)	210 (196–233)	49 (0–145)
	TP	0	NA	NA	NA
Okere Inlet	None	4	193 (133–319)	212 (200–219)	35 (1–72)
	TP	7	610 (361–1256)	327 (279–405)	159 (0–383)
Te Weta Bay	None	5	222 (178–287)	249 (232–255)	90 (0–148)
	TP	5	376 (267–576)	284 (260–335)	67 (0–329)
Southern	None	1	118	200	49
Geothermal	TP	3	310 (270–332)	266 (250–279)	166 (22–244)

Fish mass and length were taken at the time of surgery and detection duration is based on the date of tagging until the last detection was received. Mean values are presented along with minimum and maximum values observed (in parentheses) for catfish receiving transmitters with or without a temperature–pressure (TP) sensor

ensure triangulation and allow for detailed positioning of fish within the western bay of the lake, which is where catfish were first detected (Fig. 1). The other receivers were positioned strategically to determine movements of catfish into and out of Te Weta Bay (where the highest density of catfish occurred), and to determine if catfish were moving outside of the western bay, into the Eastern Basin, Okawa Bay, Okere Inlet, and the Southern Geothermal shoreline. Okere Inlet leads into the primary outlet (the Kaituna River). One receiver was positioned at the entry of the primary inflow (the Ohau Channel). Three reference tags (V16, 69 kHz, random transmission interval, 540–660 s) were centrally deployed within key areas of the receiver array to assess positioning accuracy. Individual receivers logged the fish identification number (ID), any associated sensor data from acoustically tagged fish that were detected, and a date and time stamp.

Data processing

Detections of acoustically tagged catfish were recorded by 33 of the 42 receivers deployed. Notably, three of the six receivers in the eastern portion of the lake (i.e., outside of the western basin; Fig. 1) did not record any detections and the eastern side of the lake was excluded from our analyses. Very few detections occurred on the eastern receivers that did receive catfish transmissions (0.1% of fish detections), suggesting that at the time of the study this part of the lake was not extensively used by the catfish population (Fig. S1).

Raw telemetry data were screened to remove duplicated data, positions outside of the lake boundaries, and data with horizontal positioning error > 20 (Smith 2013). This resulted in a study of 22 catfish with a total of 434,030 detections and 63,339 spatial positions. At least one and up to 20 fish were detected per day. Over half (n = 15) of the catfish retained in these data sets were not detected after April 2019, possibly due to transmitter expulsion (Sakaris et al. 2005b), such that the data from May 2019 onwards were collected from at most 7 fish. From early April to late September 2019, > 95% of daily catfish detections over 24 h occurred during the night. This is consistent with a previous telemetry study of brown bullhead catfish in a small embayment of Lake Taupō, where 84% of all detections were received from 1900 to 0700 h (Dedual 2002). Here, we focus on nighttime activity to maintain consistency in the daily timing of detections throughout the year-long study.

All following analyses were conducted with the programming language R (v. 4.4.2; R Core Team 2024). The “dplyr” (Wickham et al. 2023) and “tidync” (Sumner 2025) packages were used for data wrangling and the packages “tmap” (Tennekes 2018), “ggplot2” (Wickham 2016), and “viridis” (Garnier et al. 2024) were used for visualization. We assigned diel periods to detections and positions based on sunrise and sunset times with the function “getSunlightTimes” from the “suncalc” package (Thieurmel and Elmarhraoui 2022; latitude = −38.032433 and longitude = 176.415589). We restricted our analyses to fish observations between sunset and sunrise except where otherwise noted. This time window targeted the period when population control nets were set and can inform optimization of management protocols. Seasons were divided based on previous freshwater research on the North Island that categorized spring as September—November, summer as December–February, autumn as March–May, and winter as June–August (Stark and Phillips 2009).

Depth and temperature occupancy

To assess the typical boundaries of fundamental fish habitat metrics used by catfish, we used detection data from the tags with both pressure and temperature sensors (n = 8 after data filtering; Table 1) to examine patterns in depth and thermal occupancy over the year. We calculated individual-level mean depths and temperature values per night, along with the 2.5–97.5th percentiles at the population level. Of the 142,550 depth recordings, 40% were registered above the surface of the lake (up to −0.6 m). Given that the accuracy of pressure sensors is ± 0.5 m up to depths of 34 m (Vemco Ltd., Canada), we retained these depth readings in our analysis but converted them to 0 m and consider them to be effectively at the surface of the lake.

We modelled internal fish temperature and depth as predicted by season and lake region with generalized linear mixed models, including fish ID as a random effect (package "glmmTMB;" Brooks et al. 2017). Based on receiver locations, we assigned detections as either within the central basin, in the Southern Geothermal shoreline, in Okawa Bay, in Te Weta Bay, in Okere Inlet, or in the eastern part of the lake (Fig. 1). Depth and temperature were modelled with Tweedie and Gamma family error distributions, respectively. Models were visually assessed with residual plots via the "DHARMa" package (Hartig 2024) and predictor significance was tested with Type II Wald χ^2 tests with the "Anova" function in the package "car" (Fox and Weisberg 2019).

To understand the degree of selectivity exhibited by catfish, we examined temperature use relative to the available temperatures in the western basin of Lake Rotoiti. We defined this range to be the difference between the 10th and 90th percentile temperatures at 1 m below the surface. Water temperature data was modelled using a 3D hydrodynamic model of Lake Rotoiti, developed using AEM3D (Hydronumerics, Melbourne, Australia), first published in Prentice et al. (2026). The high-resolution model was configured and calibrated (RMSE of 0.1 °C in the main basin) at a 2-min timestep, 100 m × 100 m grid cell, and 1-m depth resolution. Surface (1 m depth) water temperature data were extracted at 12-h intervals for the duration of the study period to examine the spatial variability in thermal occupancy by catfish. Kruskal–Wallis tests were used to examine seasonal variation in the range of temperatures available to catfish.

Spatial habitat associations

We used catfish positions to calculate and describe seasonal patterns in nearshore compared to offshore habitat use. To quantify nearshore habitat use we calculated the shortest possible distance from each position to the lake outline using the function "dist2shore" from the R package "geosphere" (Hijmans 2024). Similarly to depth, we calculated individual-level daily mean distance to shore for each catfish in addition to population-level 2.5th–97.5th percentile ranges and describe this relationship over time throughout the study period. We also report the mean nightly percent nearshore positions for each season, for nights when individuals were positioned at least three times, defining nearshore as the lake area bounded by the shoreline and the 7 m isopleth (i.e., photic depth; Hamilton et al. 2010). Because the receiver array only allowed for positioning in Okere Inlet, Te Weta Bay, and the central basin, this section focuses on fish behaviour and habitat associations in this western region of Lake Rotoiti. Nineteen fish were positioned from a total of twenty-four that were detected on at least one receiver in this area.

A subset of the depth readings (n = 5602) were associated with a spatial position, which allowed us to calculate the distance off the lake bottom based on bathymetry. We matched catfish detections to bathymetric data by spatial position with the function "st_join" from the "sf" R package (Pebesma and Bivand 2023). We describe the range of lake bathymetry over which catfish were detected, along with the distribution of distances from the lake bottom where they were positioned. As bathymetric depths represent mean values across 100 m × 100 m grid cells, some pressure-sensor-derived depths (accuracy ± 0.5 m) were calculated as negative (i.e., below the lake bottom). We retained these values in subsequent analyses and considered negative positions to represent locations closely associated with benthic habitat.

We examined the relationships between bathymetry and lake bottom to describe behavioural patterns in the degree of benthic habitat use by catfish through kernel density estimates (KDE) with the R package “ks” (Duong et al. 2024). Following Richter et al. (2024), we used inverse frequency weighting such that the more positions were associated with an individual fish, the lower each position was weighed to produce the overall density estimate. We calculated KDEs for each season and for all data combined.

Movement analysis

Minimum convex polygons (MCPs) are a metric of animal space use derived through the calculation of the area of the smallest possible polygon that encompasses the observed positions (Kraft et al. 2023). To overcome issues of sample size and study duration on MCP areal estimates (Rogers and White 2007), we calculated MCPs individually by fish and by night to standardize study duration between units. We calculated MCPs using an increasing number of data points (Odum and Kuenzler 1955; Blanchfield et al. 2023) and determined that 35 positions provided a peak MCP. Therefore, we calculated 95% MCPs with a random sample of 35 data points per fish-night using the function “mcp” from the R package “adehabitatHR” (Calenge 2006). Random sampling was standardized using the “char2seed” function from the package “TeachingDemos” (Snow 2024). We compared natural log-transformed MCP areas by season with a generalized linear model using the R package “lme4” (Bates et al. 2015). Fish ID was included as a random effect and model fit and output were assessed with the “DHARMa” and “car” packages as above.

To understand rates of movement, we calculated horizontal distances between successive spatial positions using the “distHaversine” function in the “geosphere” package (Hijmans 2024). We calculated a nightly distance moved per hour to account for temporal variation in night duration over the year-long study period. Movement rates followed similar patterns when calculated with the same random sample of 35 data points per fish-night as above compared to when calculated with all available data, so we retained all data in these calculations. Because the distance calculated between each successive pair of detections is a minimum estimate (i.e. straight-line distance), the calculated movement rate is highly conservative. We report population-level patterns in horizontal distances travelled.

We modelled the relationship between MCP area (natural log-transformed), nightly horizontal distance, and day-of-year with GAMs, using the R package “mgcv” (Wood 2017). These models are effective for non-linear relationships and thus in studies of wild fish (Wood 2017; Pedersen et al. 2019). We built three models using fish ID as a random effect (to account for the non-independence of repeated measurements) and day-of-year, lateral distance, or day-of-year and horizontal distance (as separate smooths) as predictors of MCP area. We checked model fit using the function “gam.check” from “mgcv.” Comparisons of model fit, including to a null model, were based on Akaike’s information criterion (Akaike 1998; Burnham et al. 2011). We selected smoothness using the restricted maximum likelihood method (REML). Where necessary, we adjusted the number of basis dimensions (i.e., model "wiggliness;" Wood 2008) until there were no significant patterns in the residuals, while keeping a physically reasonable number of basis splines.

Site fidelity

To investigate catfish site fidelity, we examined detection sequences for each individual based on receiver location (Gabadinho et al. 2011; Lowe et al. 2020). For this analysis, we used both day and night detections but assigned each day as beginning at sunset to maintain the same data groupings across analyses. Detections were assigned to regions designated in Fig. 1. When a fish was detected in more than one of these lake regions on a given day, we took the region in which at least half of their detections occurred as their daily position (Elliott et al. 2023). We applied the last-observation-carried-forward (LOCF) method to the receiver detection data to ‘fill in’ the days when fish were not detected (McDougall et al. 2014; Colborne et al. 2019). We calculated the percent of fish-days at each of the five major regions (central basin, Southern Geothermal shoreline, Te Weta Bay, Okere

Inlet, and Okawa Bay; Fig. 1) for each individual to assess the relative proportion of time spent in the region where they were originally captured. We also calculated residency index for each fish as the number of days detected in a region (not including LOCF inferences) divided by the number of days between the first and last detections of that individual, which can range from 0–1 (Kraft et al. 2023).

Results

Depth and temperature occupancy

Catfish were observed at depths from 0 to 22.1 m over the study period (mean value = 2.5 m). Deeper observations were rare, with only one fish detected at the maximum observed depth on one night (May 19, 2019), whereas all eight of the sensor-tagged fish were detected at the lake surface (0 m) at least once over 384 distinct nights. Almost all (98%) depth readings were shallower than 10 m. In general, catfish were detected at increasing depths throughout the summer, with deepest depths observed during autumn followed by a return to shallower depths over the winter (Fig. 2a). Both season and region had a significant effect on catfish depths (Tables 2, 3).

The coldest observed catfish temperature was 8 °C (August 6, 2019) and the warmest observed catfish temperature was 29 °C (April 10, 2019). Annual catfish temperatures followed a sinusoidal pattern, with mean values at a maximum in February and a minimum in August (Fig. 2b). Catfish showed little variation in temperature occupancy, such that daily body temperatures of the tagged fish were, on average, within 2 °C of each other (Fig. 2b). Season and region both had a significant effect on catfish temperatures, but the magnitude of the temperature ranges available to catfish was not statistically different between seasons (Tables 2, 3). Because nearshore areas of the lake are the fastest to warm up and cool down, offshore areas of Lake Rotoiti provided the warmest water temperatures in autumn and winter, whereas nearshore areas were warmest in spring and summer (Fig. S2).

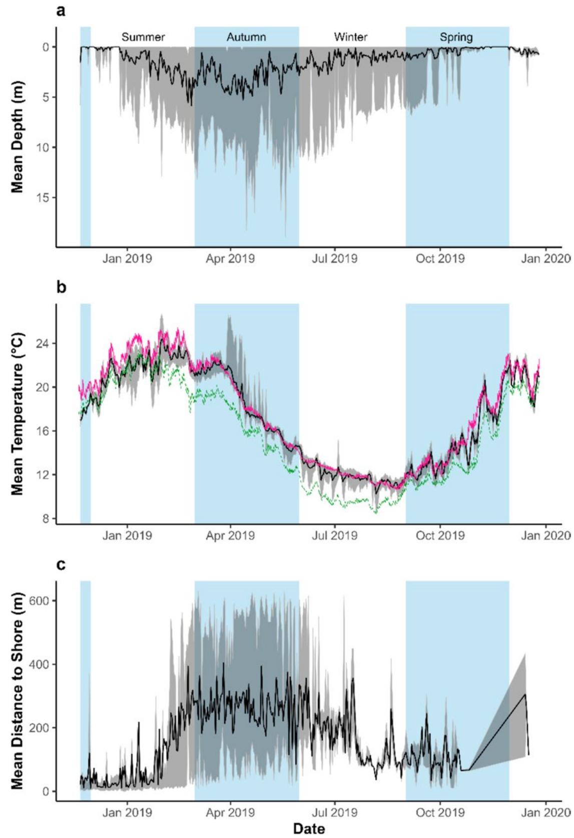

Fig. 2 Daily number of acoustically-tagged catfish detected in the western basin of Lake Rotoiti (see Fig. 1) and their mean daily observed **a** depths, **b** temperatures, and **c** distances from shore. Data are presented for nighttime detections (see Methods). Daily grand means (solid black line) are calculated as the mean of individual means, with the grey shading representing the range from the 2.5–97.5th percentiles. Catfish depth and temperature data are from the transmitter sensors, whereas distances to shore are calculated based on the minimum distance from each catfish position to the outline of Lake Rotoiti. Daily ambient temperature ranges in the western basin at 1 m below lake surface are indicated by the 90th percentile (pink line) and 10th percentile (green line) temperatures at the same depth, plotted alongside sensor-based catfish temperatures. Light blue shading indicates autumn and spring seasons

Spatial habitat associations

Catfish were positioned from 0 to 654 m from shore (mean = 117 m, interquartile range: 15–204 m). From the beginning of the study in spring (Nov 2018) until late summer (Feb 2019), nightly mean distances from shore were frequently < 50 m (Fig. 2c). Between late summer and mid-winter (mid-July 2019), catfish were routinely positioned between 200 and 400 m away

Table 2 Summary and statistical output for tests on catfish depth and temperature occupancy in Lake Rotoiti. Statistically significant results are indicated by bold text

Variable	Mean Value (range)	Statistical Test	Significance	χ^2 Value		
			Season	Region	Season	Region
Depth (m)	2.5 (0–22.1)	GLM	***p* < 0.05**	***p* < 0.05**	7455	37,616
Temperature (°C)	16.0 (8.5–29.0)	GLM	***p* < 0.05**	***p* < 0.05**	414,675	4674
Range of ambient temperatures (°C)	2.5 (0.7–5.1)	Kruskal–Wallis	$p = 0.2$		4.7	

Statistical outputs are reported for generalized linear models (GLMs) and Kruskal–Wallis tests, as indicated. In all cases, degrees of freedom = 3. Generalized linear models were run with a Tweedie family distribution for depth and a Gamma family distribution for temperature

Statistically significant results are indicated by bold text

Table 3 Mean values and ranges of catfish depth, catfish temperature, and range of temperatures used by catfish in Lake Rotoiti by season and lake region (see Fig. 1)

Variable	Category				
			Lake Region		
	Central	Southern Geothermal	Okere Inlet	Okawa Bay	Te Weta
Depth (m)	4.3 (0–22.1)	3.5 (0–18.3)	0.2 (0–5.2)	0.9 (0–7.7)	1.2 (0–13.3)
Temperature (°C)	17.0 (10.4–27.6)	15.8 (10.6–29.0)	14.9 (8.5–27.0)	21.6 (11.3–25.0)	16.5 (11.2–25.6)

Variable	Season			
	Spring	Summer	Autumn	Winter
Depth (m)	0.7 (0–13.0)	2.1 (0–13.6)	4.0 (0–22.1)	2.3 (0–16.5)
Temperature (°C)	13.6 (10.5–23.2)	21.9 (16.8–27.0)	18.0 (12.7–29.0)	11.9 (8.5–16.5)
Daily range in temperature (°C)	1.9 (0–4.9)	2.7 (0.0–6.6)	4.0 (0.5–11.3)	2.1 (0.1–4.6)

Seasons are as follows: spring is September–November, summer is December–February, autumn is March–May, and winter is June–August

from shore, followed by a return towards shore from mid-winter until the end of the study period, although this latter period was characterized by mean values close to 100 m offshore (Fig. 2c).

For nights when catfish were positioned at least three times (n = 331), we were able to calculate the percent of nearshore positions. Just under half of the western portion of the lake fulfils our definition of nearshore habitat as being bounded by the shoreline and the 7 m isopleth (41%). However, much of this nearshore habitat was within Okawa Bay, Okere Inlet, and Te Weta Bay, which had lower receiver coverage than the central basin, such that 13% of the area that produced spatial data was considered nearshore. Overall, 20% of catfish positions were nearshore. The percentage of positions in the nearshore area was greatest in spring (Sept–Nov; 83%), least in autumn (Mar–May; 5%), and intermediate in summer (Dec–Feb; 32%) and in winter (June–Aug; 35%). We observed limited nightly movement by catfish between nearshore and offshore areas, with 59% and 20% of catfish-nights containing exclusively either offshore or nearshore positions, respectively.

Catfish positions were recorded over bathymetric depths of 1–26 m, within an available depth range of 0–32 m. Overall, 4% of catfish positions were associated with bottom habitat (i.e. within 1 m of lake bottom); for the remaining 96% of positions, catfish were suspended in the water column (i.e. > 1 m from the lake bottom), with a mean depth off bottom of 7 m (median = 6 m, interquartile range = 3–10 m). For most (60%) of the suspended positions in offshore areas (i.e. lake depth > 7 m), catfish were greater than

7 m above bottom, which accounted for nearly half (48%) of the total positions.

Catfish were more often associated with the lake bottom when in nearshore areas (15% of nearshore positions) compared to when they were positioned in offshore areas (2% of offshore positions), although this pattern varied across seasons. In spring, only 2% of nearshore and no offshore positions were on the lake bottom. This increased to 41% of nearshore positions in the summer, but again catfish in offshore areas were never detected within 1 m of lake bottom (0%). More than half (60%) of nearshore positions were on bottom in autumn, compared to 2% of offshore positions. In the winter, 6% of nearshore and 1% of offshore positions were associated with the lake bottom.

Catfish positions were concentrated in the 0–5 m and 10–15 m bathymetric contour ranges (Fig. 3). In particular, shallow, nearshore positions were prevalent in all seasons other than autumn. Autumnal positions were commonly in the 10–15 m bathymetric contour range, with 25% KDEs occurring both near the bottom and in the water column at these lake depths (Fig. 3). In the summer, catfish positions were concentrated near the lake bottom in this bathymetric range, whereas in winter they were shallower in the water column (Fig. 3). Kernel density estimates for spring positions were exclusively positioned in the shallow, nearshore region of the lake, up to and including the 75%-level estimate (Fig. 3).

Movement analysis

Minimum convex polygons (MCPs) were calculated for the 12 catfish which had a sufficient number of relocations (n = 35) in at least one night (n = 254 MCPs, calculated for 169 distinct nights). Overall, MCP area ranged in size from 0 to 99 ha with a median value of 0.4 ha. Season had a significant effect on MCP area (GLM; $\chi^2 = 97.5$, $p < 0.05$). Catfish areas of use were smallest in spring (median = 0.1 ha; range: 0–32.8 ha) and summer (median = 0.1 ha; range: 0–35.7 ha), compared to winter (median = 0.8 ha; range: 0–58.3 ha) and autumn (median = 27.5 ha; range: 0–98.7 ha) (Fig. 4a).

Catfish exhibited different seasonal rates of movement (Fig. 4b). On average, individuals travelled 101 m hr^{-1} at night, with a range of 1–537 m hr^{-1} (interquartile range: 28–150 m hr^{-1}. The model that included both day of year and movement rate explained MCP size best, explaining 87% of the variation in the data. Consistent with autumn (March–May) as a period of greatest MCP area (Fig. 4a), catfish exhibited the highest rates of movement during this season (Fig. 4b). The size of MCPs generally increased with movement rate, although there were instances of high movement rate occurring

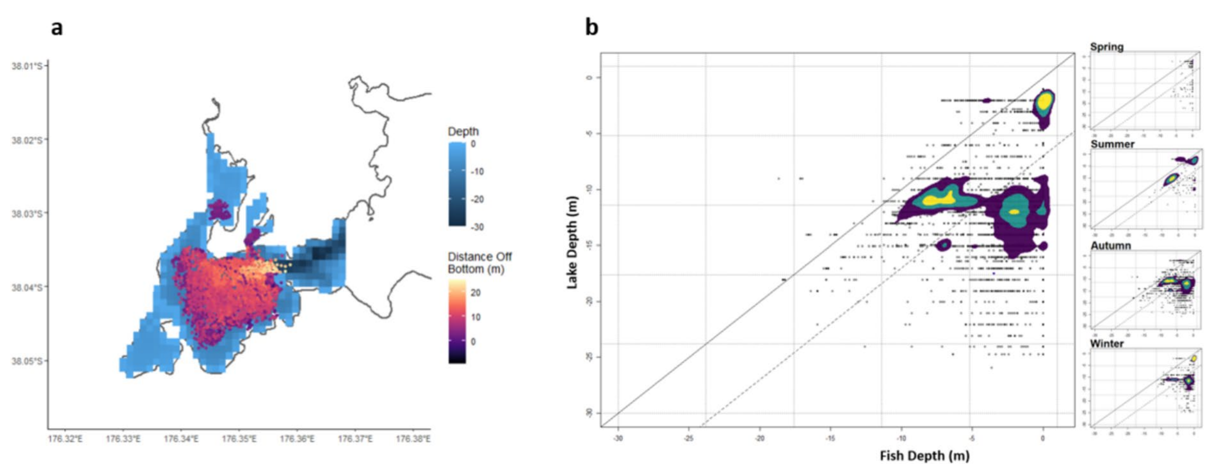

Fig. 3 Map of the western basin of Lake Rotoiti with acoustic positions of catfish shown according to depth from the bottom of the lake in panel (**a**) and the relationship between fish (sensor) depth and lake depth (bathymetry) in panel (**b**). Individual data points (grey) are shown with the 1:1 line indicated in black. The dashed line represents 7 m above lake bottom. Kernel density estimates are represented in purple at the 75% level, blue at the 50% level, and yellow at the 25% level

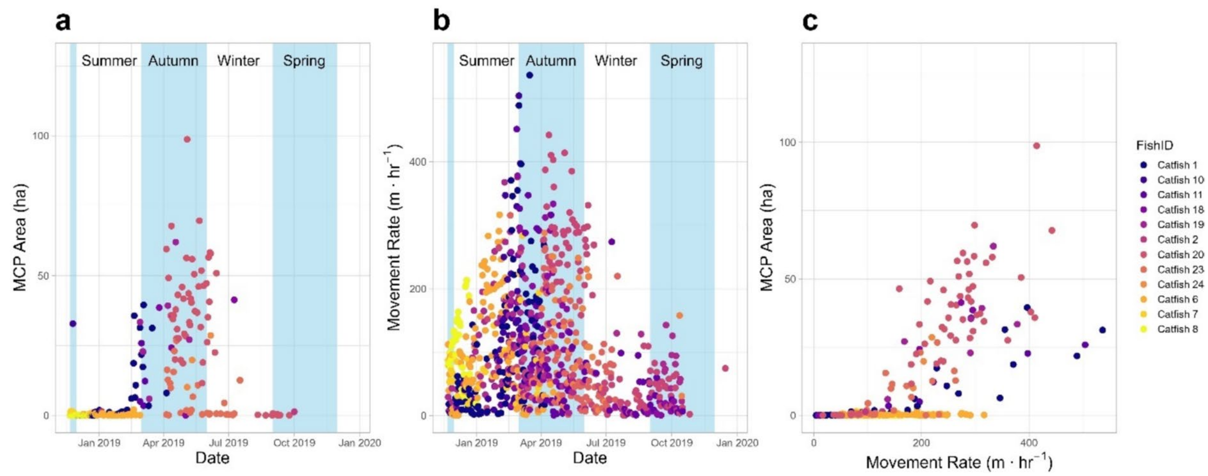

Fig. 4 Relationship between nightly horizontal distance moved by acoustically-tagged catfish per hour and area of 95% minimum convex polygons (MCPs) in Lake Rotoiti. Nightly horizontal distances and MCP size, both calculated from spatial positions, are shown across seasons (**a**, **b**) and relative to each other (**c**). Shading in panels (**a**) and (**b**) indicates approximate seasons (autumn and spring shaded). All panels have colour-coding to show individual fish ID

within a small MCP area, particularly during summer (Fig. 4c).

Site fidelity

We used sequence analysis to visualize whether acoustically tagged catfish showed evidence of site fidelity. On average, about four-fifths (79%; range: 29–100%) of daily positions were observed, while the rest were derived based on the LOCF method. Most derived sequences longer than 5 days were in Okere Inlet (n = 22), which accounted for 32% of observed daily locations and 47% derived daily locations. The majority of daily-assigned positions occurred in Okere Inlet, followed by the central basin, the Southern Geothermal shoreline, Te Weta Bay, and Okawa Bay (Table 4). This was also reflected in individual residency indexes, which were highest on average for Okere Inlet (mean = 0.2; range: 0–1) and the central basin (mean = 0.2; range: 0–0.9). While a few catfish (n = 3) showed high site fidelity and were positioned only in the area where they were originally tagged, most catfish were positioned at sites other than where they were captured and tagged (Table 4, Fig. 5). Fish tagged in all locations except Okawa Bay were most commonly in their original tagging location (Table 4). Fish tagged in Okawa Bay were most commonly in the central basin, which was also the second most common location for all other fish (Table 4). Catfish tagged in all locations were observed in the central basin and in Okere Inlet, and those tagged in all locations except for the Southern Geothermal shoreline were observed in Okawa Bay for at least 20 fish-days (Table 4). Only fish tagged in Te Weta Bay had daily positions in this part of the lake (Fig. 5).

Discussion

In their native range of North America, brown bullhead catfish are characterized as nocturnal littoral-benthic predators that show limited movement and high site fidelity (Scott and Crossman 1973). This general accepted description of catfish habitat use and behaviour is not what we observed in the recently invaded Lake Rotoiti in Aotearoa. Instead, catfish routinely occupied offshore regions, were highly mobile, and rarely associated with the lake bottom when offshore, especially in autumn when rates of nightly movement were greatest. Only in spring and summer were catfish regularly found in nearshore areas, and during these seasons they were less active but more likely to be associated with bottom habitats. Most of the tagged catfish did not show strong site fidelity and were routinely located at sites distant from their original site of capture. While catfish showed

Table 4 Daily locations by tagging site for acoustically-tagged catfish in Lake Rotoiti

Tagging location	Detected location	Number of fish-days	Percent of fish-days	Residency index
Okawa Bay	Central	158	42.6	0.4 (0.1–0.7)
	S. Geothermal	0	0	0
	Okawa Bay	118	31.8	0.2 (0–0.7)
	Okere Inlet	95	25.6	0.2 (0–0.6)
	Te Weta	0	0	0
Okere Inlet	Central	463	26.5	0.2 (0–0.4)
	S. Geothermal	23	1.3	0 (0–0.1)
	Okawa Bay	28	1.6	0 (0–0.1)
	Okere Inlet	1243	70.7	0.6 (0.4–1)
	Te Weta	0	0	0
Te Weta Bay	Central	289	31.1	0.2 (0–0.4)
	S. Geothermal	152	16.4	0.1 (0–0.3)
	Okawa Bay	21	2.3	0 (0–0.1)
	Okere Inlet	3	0.3	0 (0–0)
	Te Weta	463	49.9	0.6 (0.1–1)
Southern Geothermal	Central	212	26.4	0.3 (0–0.9)
	S. Geothermal	587	73.1	0.5 (0.1–0.7)
	Okawa bay	0	0	0
	Okere inlet	4	0.5	0 (0–0)
	Te Weta	0	0	0

Locations assigned based on receiver locations for which each catfish was detected each day. Residency index values are presented as means (min–max). Ranges are presented when all values round to 0 if there were any values > 0 for that tagging location-detected location combination

more expansive movement patterns than originally expected, we recorded very few detections (< 0.1%) outside of the western basin of Lake Rotoiti, suggesting there may be barriers to dispersal.

Thermal rigidity

Temperature has a strong influence on the habitat choices of fishes (Magnuson et al. 1979). Acoustically tagged catfish occupied a narrow range of water temperatures (± ~ 2 °C) on a daily basis and appeared to target the warmest water available. Catfish body temperatures closely followed the maximum daily water temperature available in the western basin throughout much of the year. In spring and summer, when the nearshore areas of the western basin of Lake Rotoiti are warmer than offshore areas, catfish were found in nearshore regions. As the lake cooled in late summer through autumn, this trend was reversed and the warmest waters were found in the offshore areas (Fig. S2). This cooling period appeared to trigger a change in catfish behaviour and habitat use, whereby tagged fish moved further from shore, were not associated with benthic habitat, and greatly increased movement rates and nightly areas of use (minimum convex polygon). However, their movement was largely constrained to the western basin. It is possible that cooler water temperatures in the much larger eastern basin acted as a barrier to catfish expansion. Thermal acclimation experiments have shown that catfish prefer warmer water than their acclimation temperatures (Richards and Ibara 1978), which the authors suggest could explain the movement of catfish to the thermal discharge canal of a power plant during autumn, effectively creating a "thermal trap". Similarly, non-native wels catfish (*Silurus glanis*) avoided a colder (~ 3.6 °C) tributary at the confluence of a large Italian river (Nyqvist et al. 2022). The warmer, shallower western basin of Lake Rotoiti may act like a "temperature trap" that is effectively slowing the expansion of catfish into the larger eastern basin of Lake Rotoiti. In support of this idea, monitoring data from 2016 to 2018 showed few catches of catfish in the eastern basin by 2018, around the time of this study, despite a substantial increase in catfish abundance in the western basin during this same period (Francis 2019).

Temperature occupancy also has implications for catfish feeding rates. In the northern part of its native

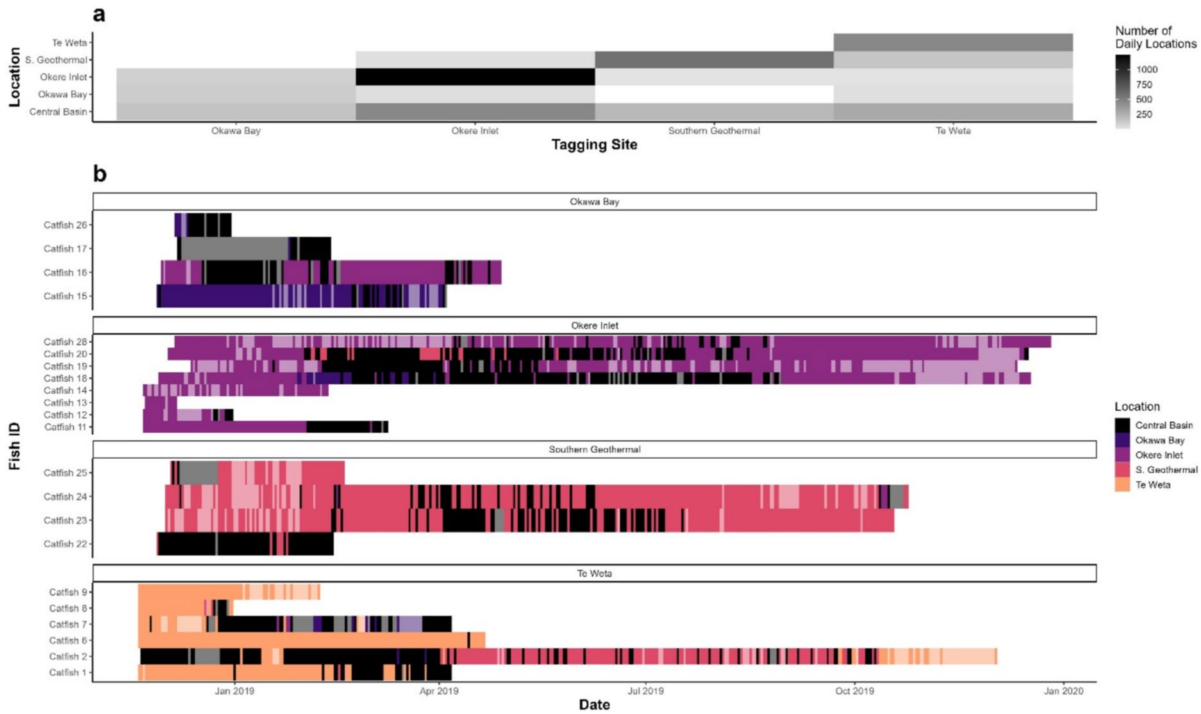

Fig. 5 Sequence analysis of acoustically tagged catfish in the western basin of Lake Rotoiti based on telemetry receiver locations (Fig. 1). The daily location to which a catfish was assigned is based on the majority of positions on a given day for that individual. Panel **a** shows the number of fish-days per location by tagging site. Panel **b** shows the location sequences through time. Derived regional positions are indicated by the lighter shading and are based on the last observation carried forward. Sequences are facetted by the tagging location for each fish

range, catfish are known to reduce feeding during winter, when water temperatures cool to 4 °C or less (Keast 1968). In the western basin of Lake Rotoiti, catfish spent much of the cooler winter months at temperatures closer to 12 °C, with 8 °C as the coldest recorded body temperature of a tagged catfish. Thus, catfish in Lake Rotoiti do not experience the temperature extremes of those in northern hemisphere temperate lakes, and, therefore, have the potential to feed and grow year-round. Laboratory studies and bioenergetics modelling indicate the highest feeding and growth rates by catfish at warm temperatures (~ 20–30 °C; Keast 1985; Hartman 2017), which were available in Lake Rotoiti for roughly half the year (Nov–April). Based on the temperatures occupied by tagged catfish, regular feeding is expected in Lake Rotoiti during autumn and winter, although at lower rates than in summer (Keast 1985). Interestingly, after accounting for climate effects, populations of brown bullhead catfish were found to have similar growth rates in their native and invaded ranges (Rypel 2013).

Benthic plasticity

Acoustically tagged brown bullhead catfish tended to occupy the upper waters of Lake Rotoiti, especially over the spring–summer period, and this occurred in both nearshore and offshore habitats. Over the year-long study, catfish depth averaged 2.5 m with most detections (98%) at depths of 10 m or less. This range of depth use is similar to what was observed for invasive catfish in a small embayment of nearby Lake Taupō, although the maximum observed catfish depth in Lake Rotoiti (22 m) was deeper than that observed for Lake Taupō (17 m; Dedual 2002). Likewise, seasonal patterns of depth use in Lake Rotoiti were similar to what Dedual (2002) observed for catfish in Lake Taupō, with catfish being shallowest in spring and summer and deeper in autumn and winter.

Spatial positioning telemetry allowed us to determine the location of tagged catfish and subsequently determine their seasonal presence in nearshore areas and their relationship to benthic habitat, all of which was not possible in the earlier study in Lake Taupō. In contrast to previous research, we found very little association with the nearshore region and the lake bottom by catfish, a species that is regarded as a littoral-benthic dweller in its native range (Scott and Crossman 1973). A total of 20% of catfish positions were found in the nearshore, defined as the area delineated by the 7 m isopleth and the edge of the lake. We estimated that the nearshore area accounted for 13% of the spatial coverage of the receiver array in the western basin and therefore although catfish showed a slight preference for the nearshore area, it was much less than expected. We chose the 7 m isopleth as an outer boundary to define the nearshore area as this is the maximum depth at which fyke nets are routinely set to eradicate catfish from the lake (Francis 2019). This depth is also consistent with findings from catfish in their native range, which occupy depths of up to 8 m depth and were only captured in littoral nets, not pelagic sets, in late summer (Stables and Thomas 1992).

Another striking finding was that invasive catfish in Lake Rotoiti were rarely associated with the lake bottom. Only 4% of all positions were considered benthic (within 1 m of the lake bottom) with the remaining 96% of positions averaging 7 m from the lake bottom. With 80% of positions in the offshore area and 60% of those > 7 m off bottom, pelagic habitat was the most commonly occupied habitat by invasive catfish in Lake Rotoiti. Invasive brown bullhead catfish in a Polish lake were captured in both pelagic and littoral areas in summer, to an estimated maximum depth of 9 m in pelagic areas (Ulikowski et al. 2022). Low dissolved oxygen likely limited the use of deeper habitats in this study (Ulikowski et al. 2022). In addition, observations of large schools of pelagic catfish near drop-offs have been observed in Lake Taupō (Dedual 2002). Collectively, these studies reveal that pelagic habitat use is a common feature of invasion behaviour by catfish in newly invaded lakes, which deviates from the strong littoral and benthic associations they are known for in their native range.

Seasonal activity

Catfish showed distinct seasonal patterns of activity in the western basin of Lake Rotoiti, with a peak period of movement and area occupied in the autumn when fish moved offshore. The peak in autumn activity was particularly striking, with large median nighttime areas of use (i.e., MCP area; 27.5 ha) roughly 300-fold higher than those in spring and summer (0.1 ha). Larger areas were used in winter than in spring and summer, but the largest nightly areas of use (~ 100 ha) occurred during autumn, coinciding with the maximum nightly movement rate of nearly 600 m·hr^{-1}. There are limited seasonal spatial and movement data available from catfish in their native range to compare with those of invasive catfish. Catfish areas of use in their native range (Anacostia River, USA), were least in summer (mean minimum polygon area = 4.5 ha), and roughly four-fold smaller compared to the spring and the fall-winter periods (Sakaris et al 2005a). While the absolute sizes of areal use estimates are not directly comparable between studies due to different methodologies and systems (i.e., manual tracking over an entire season in a river; Sakaris et al. 2005a), the relative differences among seasons suggest that the marked change in habitat use and activity in autumn by invasive catfish in Lake Rotoiti represents a novel behaviour. Autumn marks the time when catfish are free from parental care duties, which coincides with warmer water temperatures now available in the offshore relative to nearshore regions, resulting in the potential exploration of large areas of the lake.

Individual variation in behaviour is commonly observed in detailed studies of animal movement. For example, rainbow trout (*Oncorhynchus mykiss*) released from a commercial aquaculture operation into a novel environment, Lake Huron, showed a diverse range of exploratory behaviour, with some individuals remaining at the release site while others were captured up to 350 km away (Patterson and Blanchfield 2013). Invasive catfish in Lake Rotoiti showed high individual variation in movement rates and behaviour. Most notably, we observed that rates of movement were not always correlated with areas of use (MCP). In many instances high movement rates were constrained to small areas. Often this occurred in the spring and early summer, which was likely associated with the reproductive period for catfish, estimated to be between September and December in Lake Taupō (Barnes and Hicks 2003). However, we also observed extensive variation within and among individuals for the relationship between movement and MCP in autumn, when catfish were most active. Site fidelity analyses also revealed high inter-individual differences in exploratory behaviour by invasive catfish. Some catfish explored much of the western basin of Lake Rotoiti, while others remained at the site of original capture for the duration of the study. Individual differences in exploratory behaviour by invasive aquatic biota have been linked to personality traits (Cote et al. 2010; Galib et al. 2022). Relative to range-core populations, range-edge populations of species undergoing range expansion can differ in dispersal tendency or ability, aggression, boldness, activity, and sociability (Pyper et al. 2024). Because of this, an invasive population can develop a continually accelerating dispersal rate (Hudina et al. 2014). Personality traits of invasive catfish were not examined in this study but collectively the movement, areal use and site fidelity analyses, support the existence of consistent individual differences in exploratory behaviour in Lake Rotoiti.

Management implications

Contrary to typical conceptions of brown bullhead catfish within their native range, broadly we found invasive catfish had high mobility and extensively occupied offshore areas, where they were often suspended above the lake bottom. Efforts to reduce catfish in Lake Rotoiti via annual culls primarily rely on the use of fyke nets set in the shallow littoral regions of the lake, at depths of 7 m or less (Francis 2019). Our spatial data on acoustically tagged catfish reveal that this management approach will successfully target fish primarily in spring and summer when they occupy this region of the lake. Over the year-long study, only 20% of catfish locations were in the nearshore region, meaning that 80% of the time, tagged adult catfish were not in locations targeted by the capture gear. Moreover, spatial analyses showed that on ~ 60% of nights, catfish remained exclusively in the offshore region and did not move into the nearshore areas. Thus, trapping efforts may be more productive if shifted offshore in autumn, and if pelagic nets were deployed. Catfish are most active and have the largest nightly areas of use in autumn. Based on previous research showing that highly active fish are more susceptible to capture (Biro et al. 2007), this may be an important season to target fish in the offshore region. It is possible that warm water attractants near trapping areas may increase catches, as may targeting areas of warm water inputs, such as those associated with geothermal springs (e.g., Tikitere Geothermal Field adjacent to the study lake). While our analysis did not focus on aggregation behaviour, a recent telemetry study has shown that the European catfish (*Silurus glanis*) can form overwintering aggregations that last for ~ 2 months (Westrelin et al. 2023). Should invasive catfish form large seasonal schools in Lake Rotoiti, as has been observed in Lake Taupō (Dedual 2002), this may be another possible option for targeted removals.

A key concern is that invasive catfish are negatively impacting kōura (native freshwater crayfish) populations (Kusabs et al. 2026), which are a culturally treasured species and important food resource that is harvested using traditional practices in Lake Rotoiti (Kusabs and Quinn 2009; Kusabs et al. 2015; Francis 2019). Our data suggest that at the time of this study, there was limited potential for habitat overlap between catfish and kōura, as only 4% of all catfish spatial positions were estimated to be on the lake bottom. The exception is during spring and early summer, when catfish are more commonly associated with the nearshore environment. It is worth noting that juvenile crayfish, which are most prone to predation by catfish, historically were most often found at shallow depths (< 10 m) in Lake Rotoiti (Devcich 1979), similar to the depths we most commonly detected catfish in this study. Because both catfish and kōura are active at night this is the most likely time when predator–prey encounters would occur.

Finally, our results highlight the importance of temperature in shaping seasonal habitat selection by invasive catfish in Lake Rotoiti, and suggest that similar lakes across Aotearoa provide conditions that would allow catfish to thrive. Notably, there is a wide thermal envelope for optimal catfish growth that lasts for roughly half of the year. This broad thermal window would allow for repeated spawning events in a single season, which could lead to rapid population growth for this species (Scott and Crossman 1973). In addition, lakes in the north island of Aotearoa remain relatively warm compared with the northern extent of the species' native range, supporting year-round growth by catfish. Consistent with this notion, we observed catfish to occupy the warmest available water throughout the year. The close association we observed between catfish body temperature and maximum lake temperature supports the use of climate-based temperature projections to assess invasion risk

(Lee et al. 2025). As climate change increasingly favours thermally tolerant, omnivorous fishes, species like catfish will almost inevitably pose a growing management challenge (Lee et al. 2025). This telemetry study has revealed several new and unexpected findings following a recent invasion, which we anticipate will help inform more effective management of catfish in Lake Rotoiti and other vulnerable systems.

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Author contributions BH, SG, CB and PW conceived and designed the field study. SG conducted the fieldwork. PJB, CM, SS, FL, and RH conceptualized research questions. THR led the data analysis and visualization. SF-D and MJP provided data support. PJB, THR, CM, and RH drafted the manuscript. All authors contributed to subsequent drafts and approved the final manuscript.

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Data availability Data are available upon reasonable request to the corresponding author.

Code availability All code used to process and analyse data are availablein the following GitHub repository: https://github.com/tazirodrigues/rotoiti-catfish-telemetry

Declarations

Competing interest None.

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